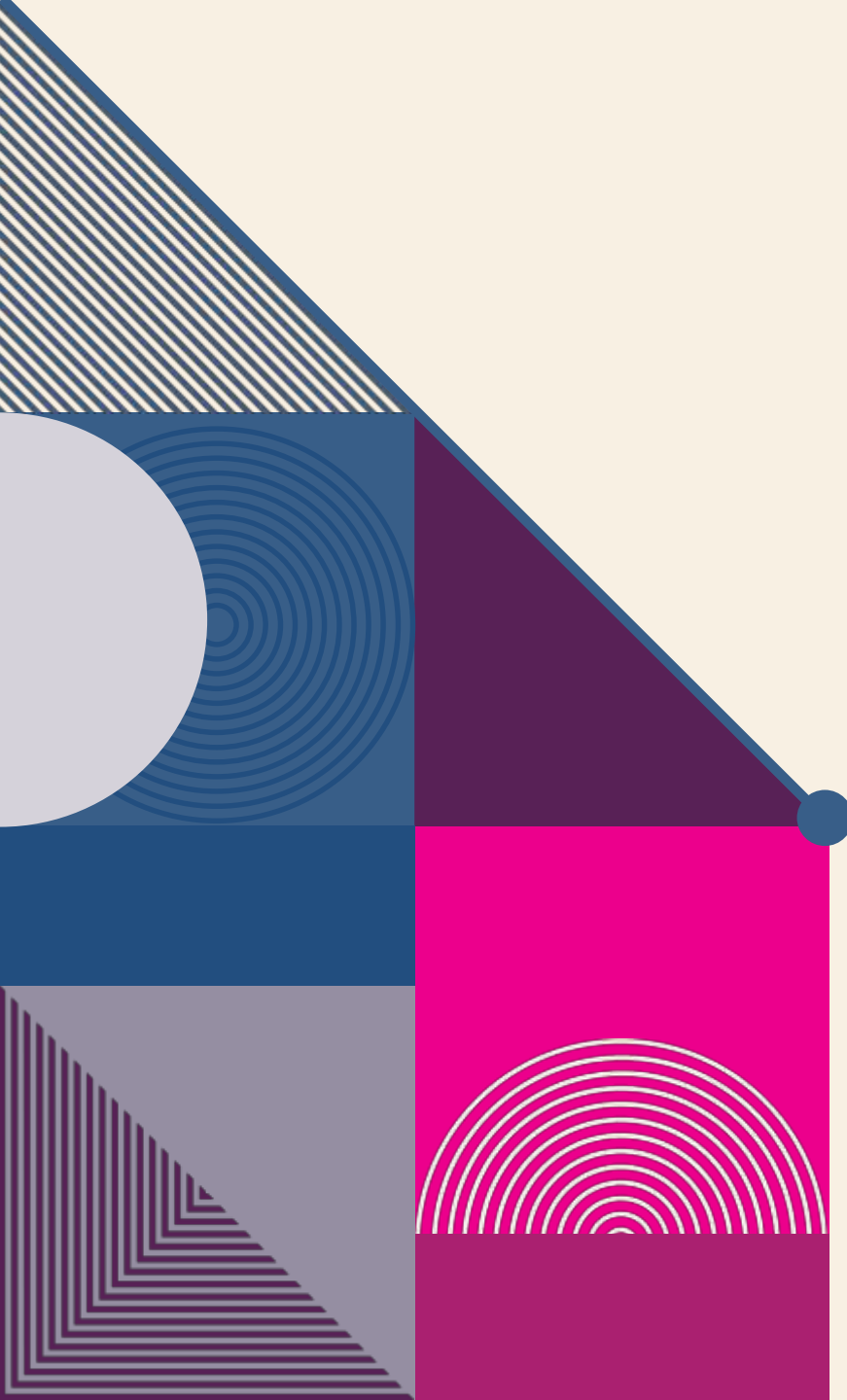
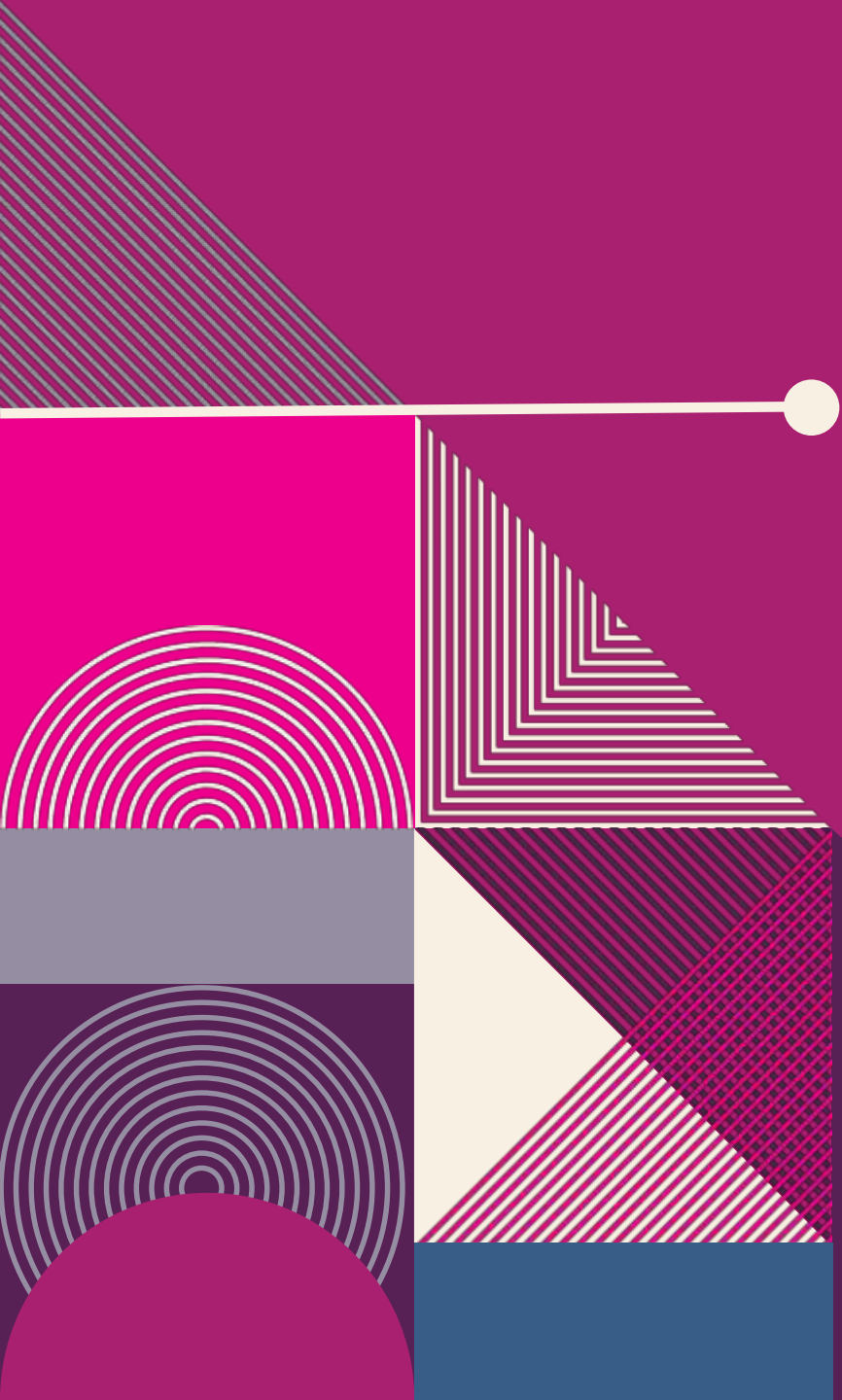


A decorative geometric pattern in the top-left corner of the slide. It consists of several overlapping shapes: a dark blue triangle with white concentric circles, a light blue square with white concentric circles, a dark blue square with white concentric circles, a light blue square with white concentric circles, and a dark blue square with white concentric circles. The pattern is composed of various geometric shapes and colors, including blue, purple, and pink, creating a complex, layered effect.

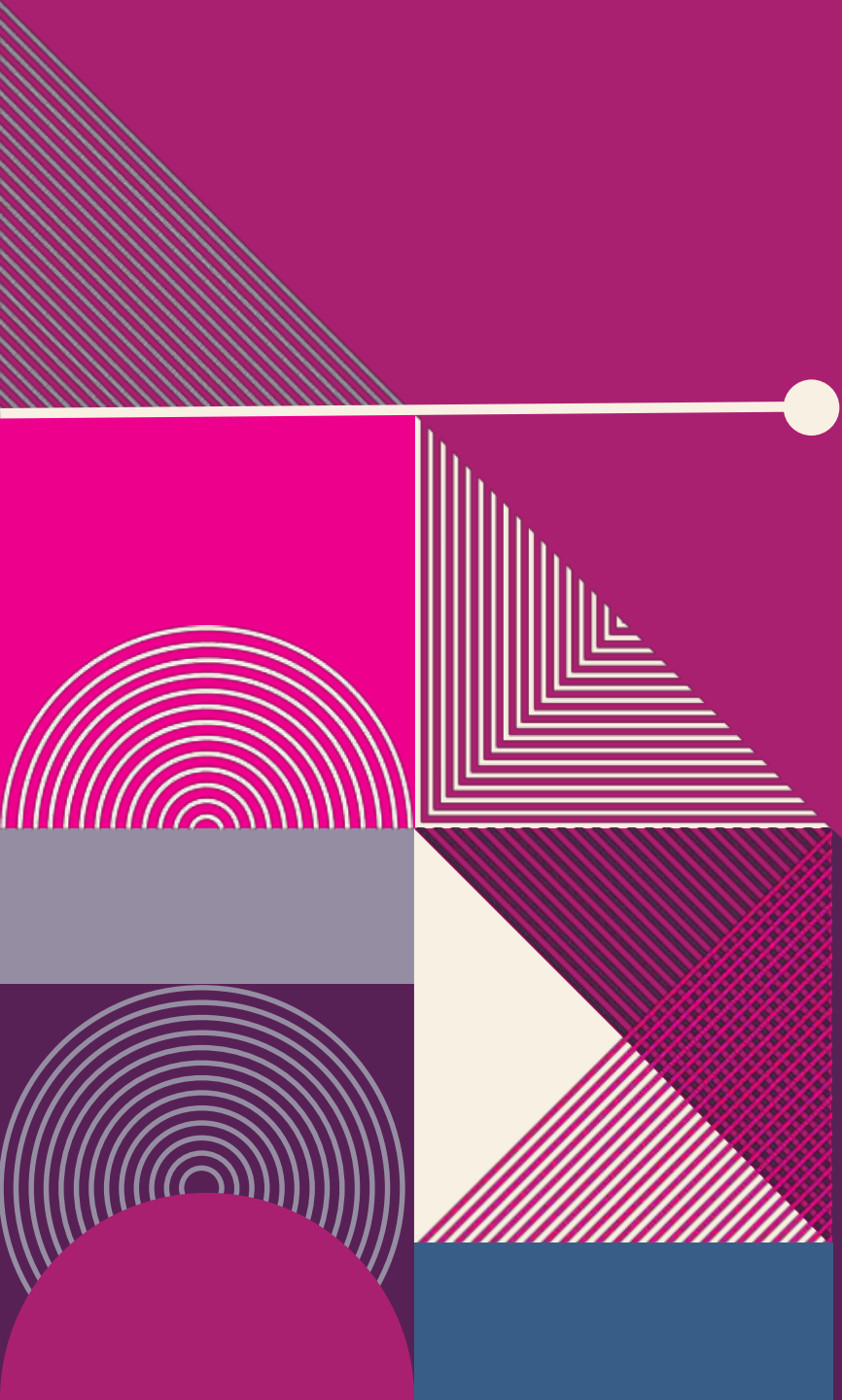
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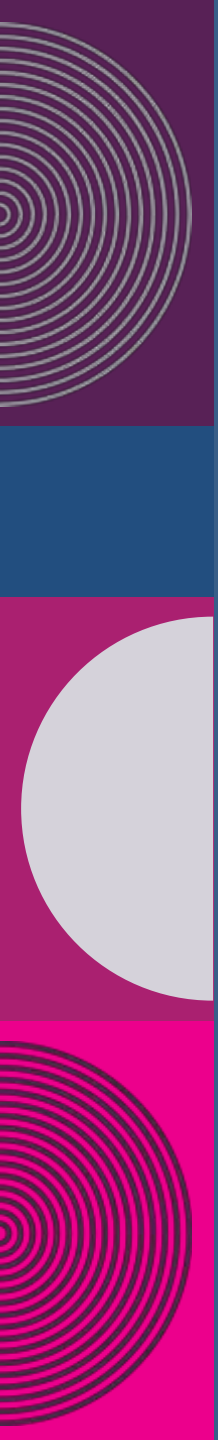
An abstract geometric design on the left side of the slide. It features a vertical stack of colored squares: a bright pink square at the top, a light grey square, a dark purple square, and a blue square at the bottom. The pink square contains a series of concentric white semi-circles. The dark purple square contains a series of concentric white circles. The blue square contains a large white semi-circle. To the right of these squares is a large white triangle pointing downwards. Above the pink square, there is a horizontal white line with a small white circle at its right end. The background of the slide is a solid magenta color.

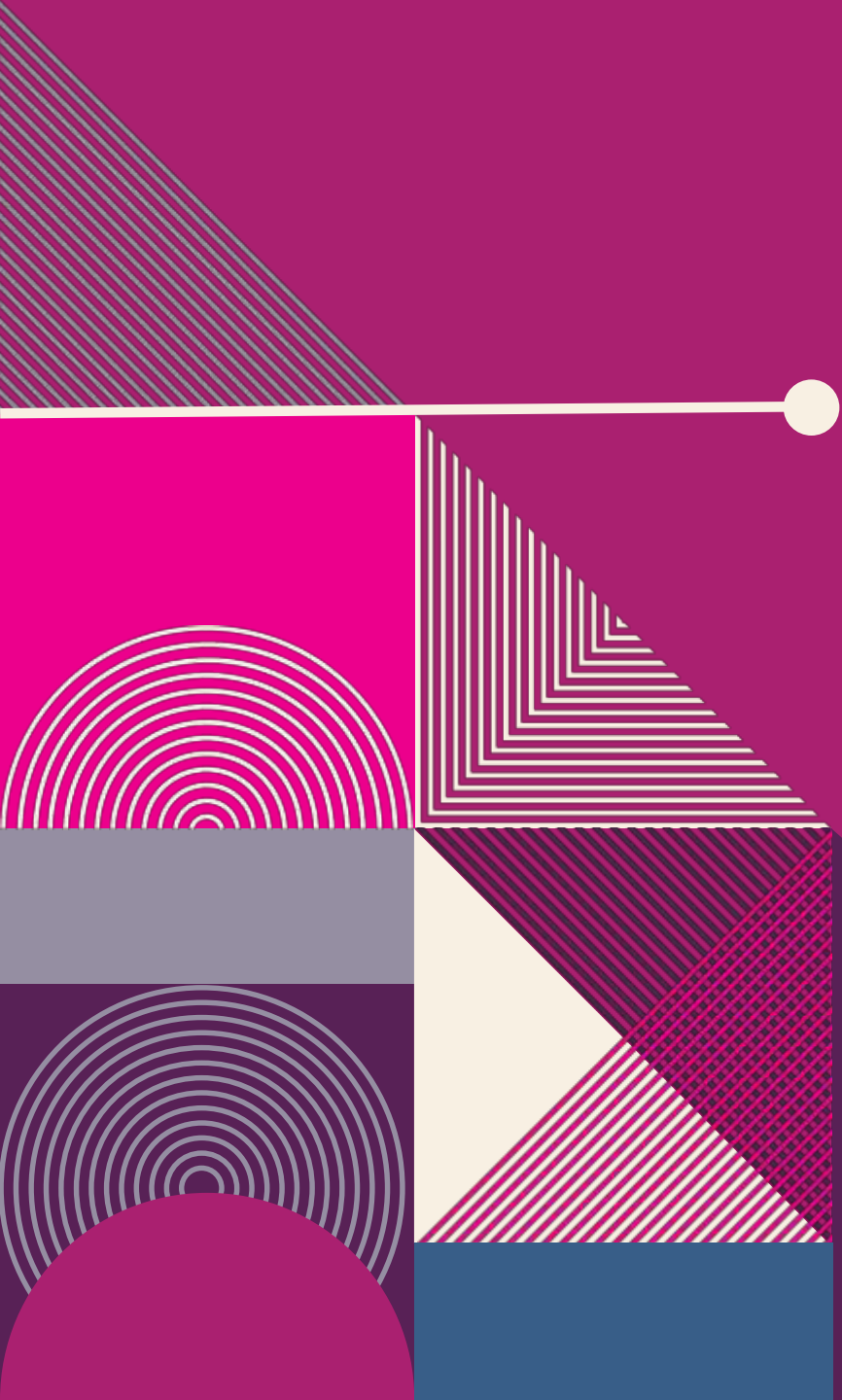
WHAT IS SMILES?

- 
- It is the abbreviation of Simplified Molecular Input Line Entry System.
 - It is a line notation used for representing molecules and reactions.
 - The purpose of SMILES is to convert the structure into a linear representation of a molecule so that computer program can understand the structure.



WHY DO WE NEED IT?

- 
- SMILES makes it easier to store and transmit chemical information of complex molecular structures electronically.
 - The syntax of SMILES is relatively simple and easy to learn.
 - SMILES is often used as input for machine learning models, facilitating tasks such as property prediction, substructure search and compound optimization.
 - Cheminformatics toolkits like RDKit and OpenBabel can convert SMILES strings back into graphical representations of molecules, making it easy to visualize and verify chemical structures.



SPECIFICATION RULES

RULE-1 (ATOMS)

C	Methane
N	Ammonia
O	Water
Cl	Hydrochloric acid
[Au]	Gold
[S]	Sulphur
[Fe⁺⁺]	Iron(II) cation
[OH⁻]	Hydroxide ion
[NH₄⁺]	Ammonium cation

- All atoms are represented by their atomic symbol as provided in periodic table.
- Elements in the organic subset { B,C,N,O,P,S,F,Cl,Br,I } can be written without brackets while other elements and ions are enclosed by brackets[].
- Within brackets, attached hydrogens and formal charges must always be mentioned.
- Hydrogen is implicitly implied in absence of bracket.
- Aromatic atoms are represented by lowercase letters.

[Na+].[Cl-]	Sodium Chloride
[K+].[C-]#N	Potassium Cyanide
[Sc]	Scandium
Sc	Methanethiol

- Ionic compounds are always represented as disconnected structures. Eg. NaCl,KCN.
- One should be aware of elements with 2 letters that could be misinterpreted by the computer.
- One such example is Scandium(Sc).
- If not mentioned in brackets, the computer can interpret that it is Sulphur attached to aromatic carbon by single bond.

RULE-2(BONDS)

- Single, double, triple, quadruple and aromatic bonds are represented by the symbols - , = , #, \$ and : respectively.
- Out of these, single and aromatic bonds are often omitted.
- Also, aromatic carbons are denoted in lowercase.
- Few examples are listed below:

CC	Ethane
CCCCI	Propyl chloride
O=C=O	Carbon dioxide
C#N	Hydrogen cyanide
C=CCC=CCO	6-Hydroxy-1,4-hexadiene
[Ga+]\$(As-]	Gallium Arsenide
c1ccccc1 (or) C1=CC=CC=C1	Benzene

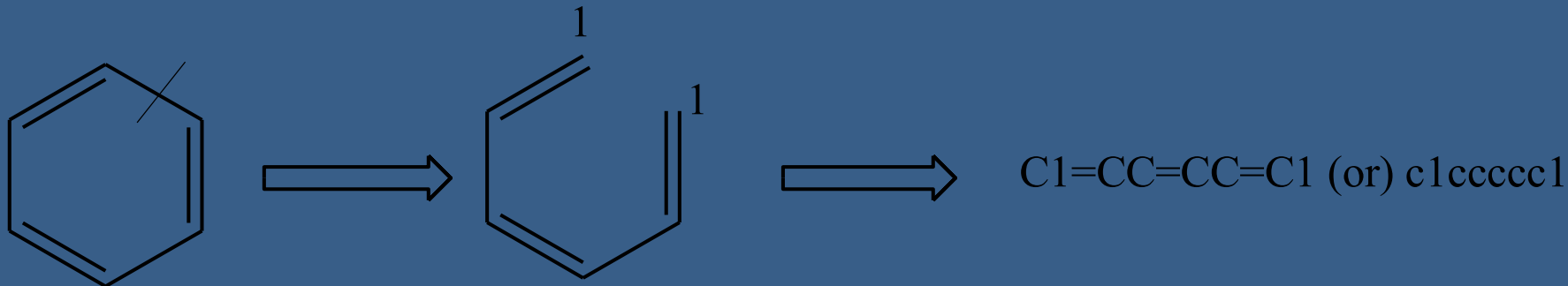
RULE-3(BRANCHES)

- Branches are specified by mentioning the group inside a parentheses.
- The group mentioned in the parentheses are always connected to the atom mentioned to the left in the string.
- Few examples are listed below:

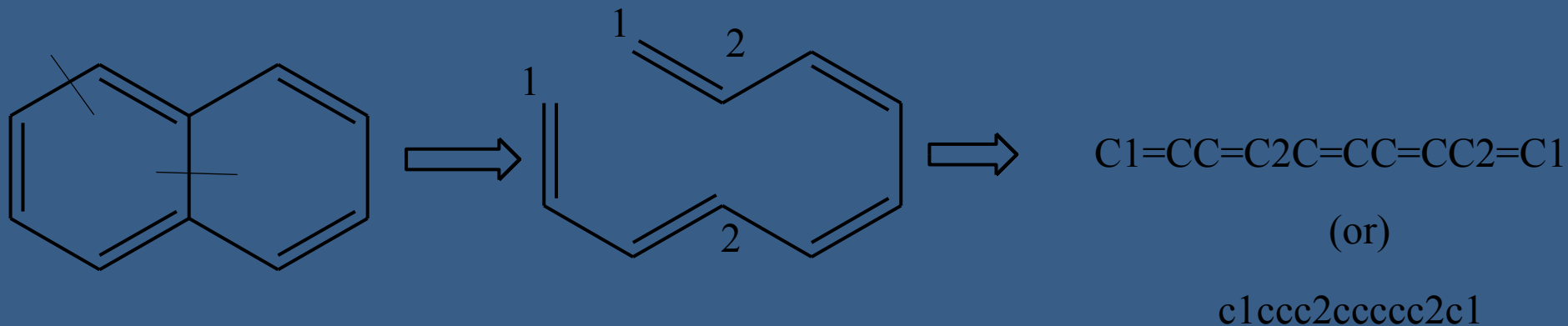
CC(=O)O	Ethanoic acid
CCC=O	Propanal
OS(=O)(=O)O	Sulphuric acid
CCC(CC)CC	3-Ethylpentane
CCCC(N)=O	Butanamide
CC(=O)CC(=O)C	Acetyl acetone

RULE-4(CYCLIC STRUCTURES)

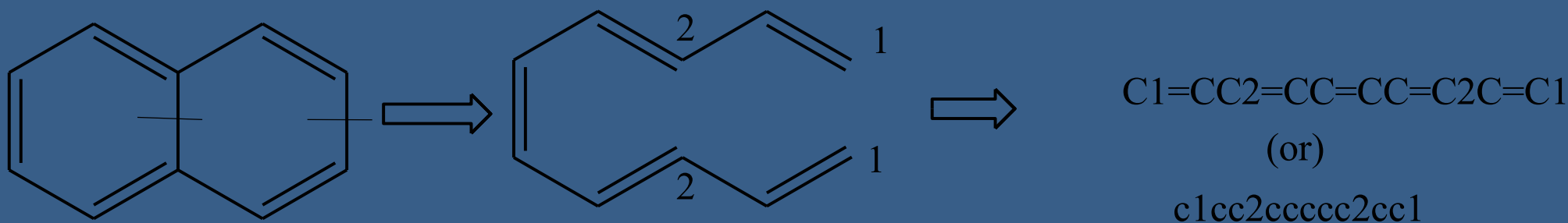
- One have to break a single bond or double bond in order to convert into a SMILES string.
- The carbons involved in ring opening/closure must be designated with same number.
- Generally, there are different but equally valid SMILES for a same structure.
- An example is mentioned below:



- The SMILES string generated completely depends on where one breaks the bond. Different ring openings/closures gives different SMILES.

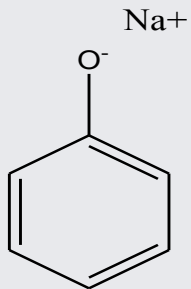


- But, if I break the bond at a different position, we get



RULE-5(DISCONNECTED STRUCTURES)

- Ions are generally written as individual structures separated by a period “.”.

<chem>c1ccccc1[O-].[Na+]</chem>	Sodium Phenoxide	
<chem>[Li+].[N+](=O)([O-])[O-]</chem>	Lithium Nitrate	<chem>LiNO3</chem>
<chem>[K+].[K+].[O-][S]([O-])(=O)=O</chem>	Potassium Sulphate	<chem>K2SO4</chem>

ISOTOPES IN SMILES

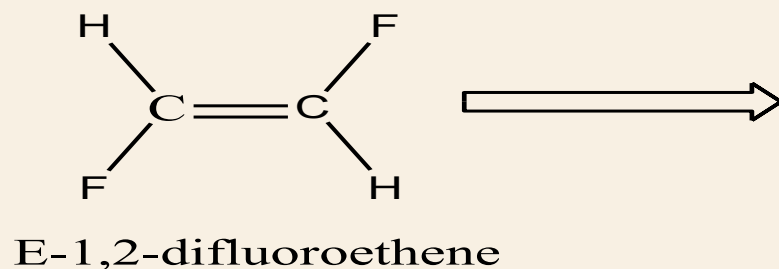
- Isotopic number must be indicated before atomic symbol of the element inside brackets.

$[^{13}\text{C}] \Rightarrow$ Carbon-13

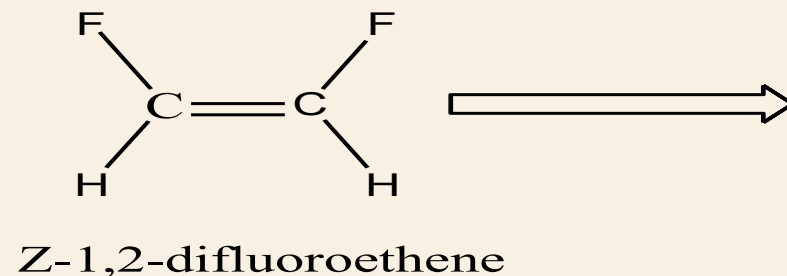
$[^{13}\text{CH}_4] \Rightarrow$ C-13 methane

CONFIGURATIONS AROUND DOUBLE BOND

- Configurations around double bond are denoted by “Directional bonds”(/ and \).



$\text{F}/\text{C}=\text{C}/\text{F}$
(or)
 $\text{F}\backslash\text{C}=\text{C}\backslash\text{F}$



$\text{F}/\text{C}=\text{C}\backslash\text{F}$
(or)
 $\text{F}\backslash\text{C}=\text{C}/\text{F}$

CHIRALITY

- To understand this, let's look into the following example



- Here, we have 4 different neighbor groups arranged around the chiral carbon atom. Looking from amino N to chiral C, the other 3 groups can either be arranged clockwise (@) or anti-clockwise(@@).

DIFFERENT CHIRAL CENTERS

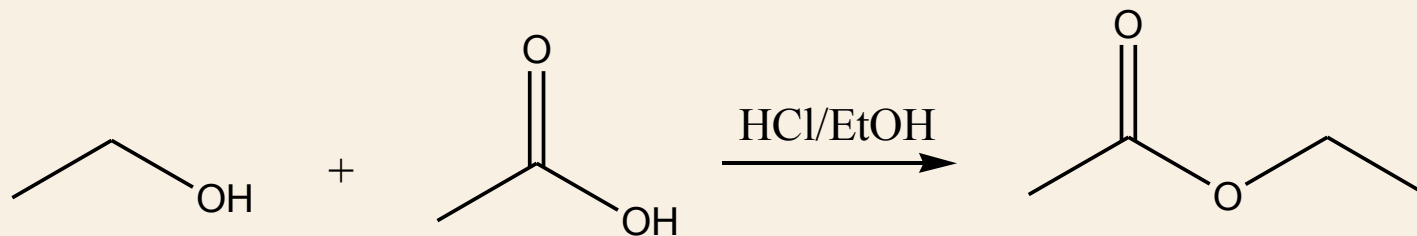
Tetrahedral	@TH1 —→ Clockwise @TH2 —→ Anti-clockwise
Allene-like	@AL1 —→ Clockwise @AL2 —→ Anti-clockwise
Square-Planar	@SP1 —→ U-shape @SP2 —→ 4-shape @SP3 —→ Z-shape
Trigonal-bipyramidal	@TB1 —→ Clockwise @TB2 —→ Anti-clockwise
Octahedral	@OH1 —→ Clockwise @OH2 —→ Anti-clockwise

REACTION SMILES

- We can also write SMILES for chemical reactions.
- The syntax contains 3 parts each separated by ">" symbol.

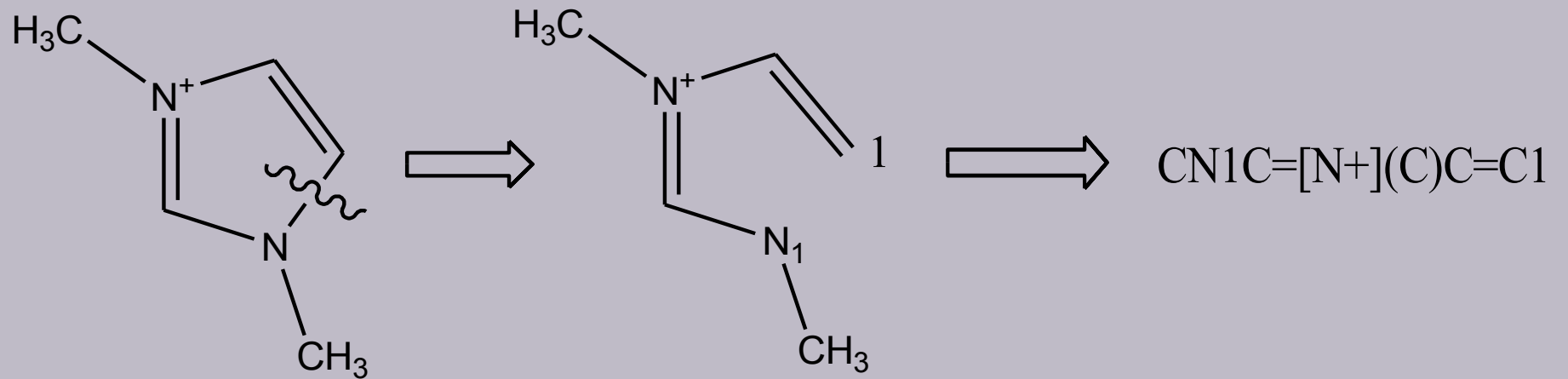
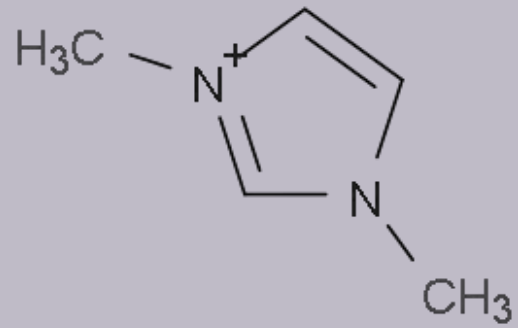
Reactants > Reagents > Products

- In each part, different molecules are separated by period ".".



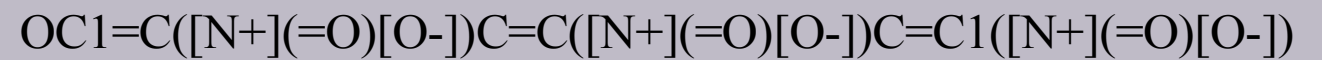
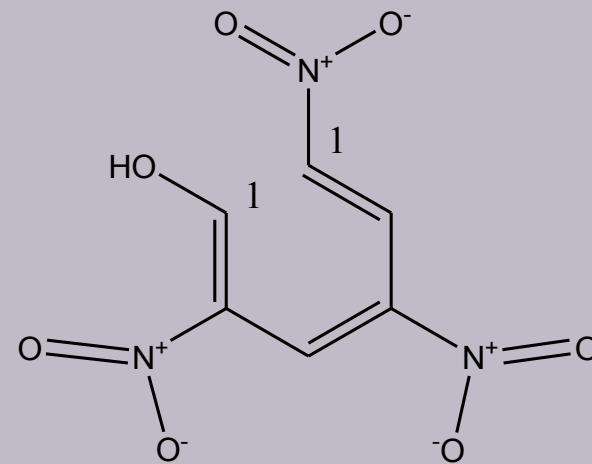
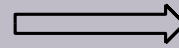
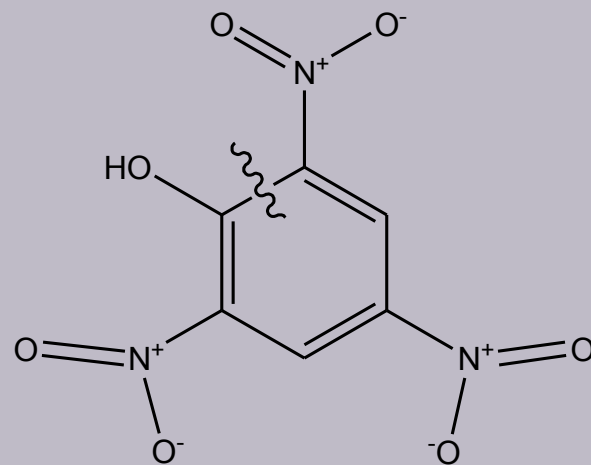
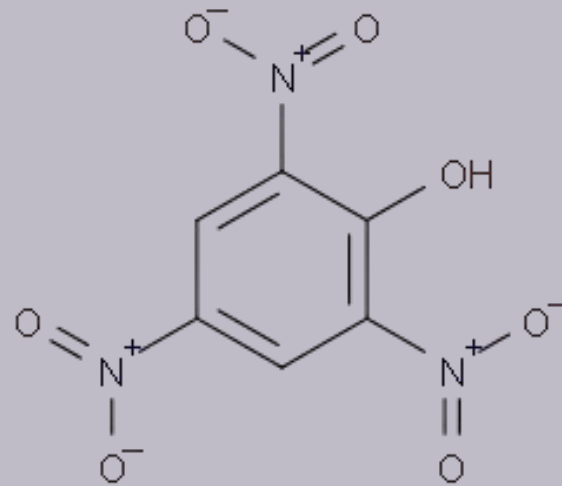
- The SMILES for this reaction is: cc(=O)O.OCC>[H+].[Cl-].OCC>cc(=O)OCC

1)



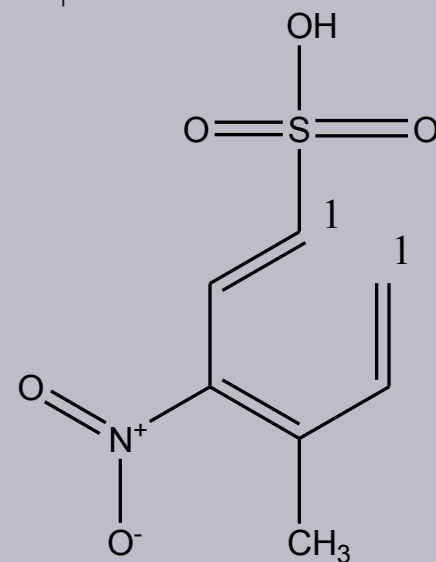
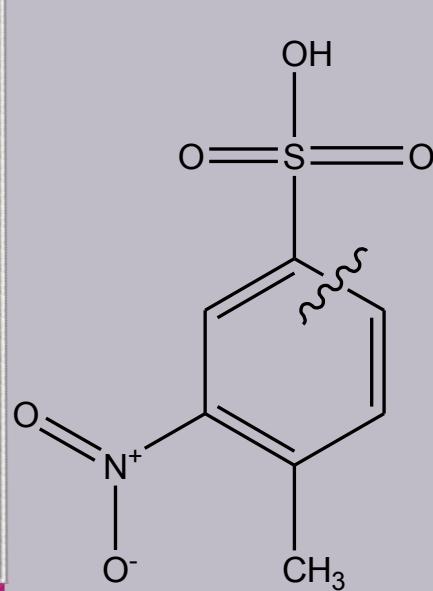
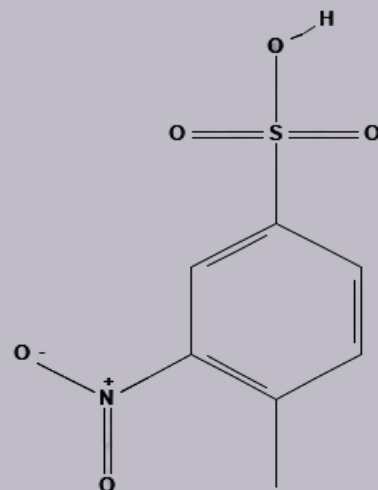
LET'S SOLVE

2)



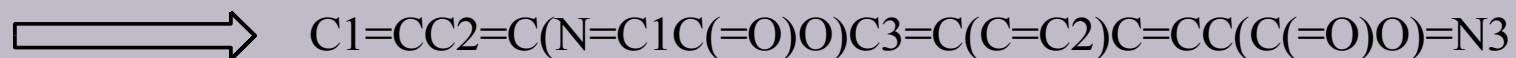
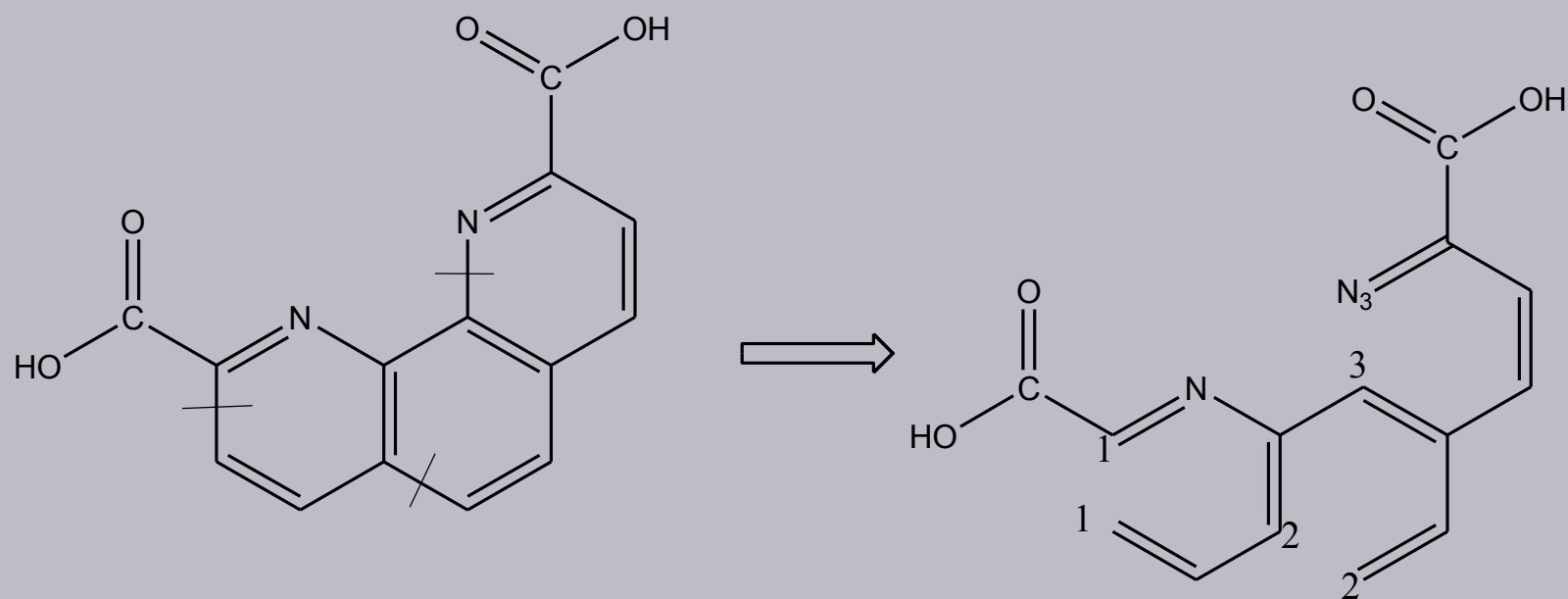
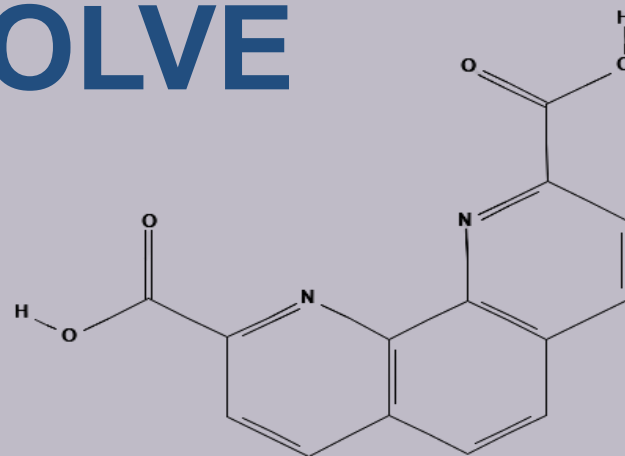
LET'S SOLVE

3)



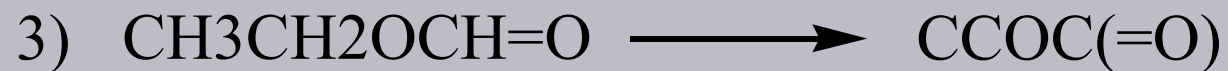
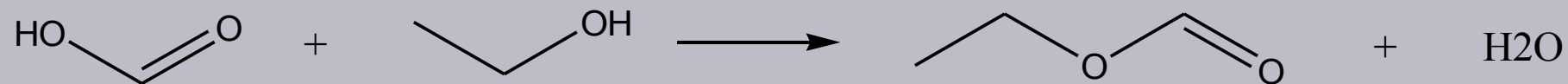
LET'S SOLVE

4)



LET'S SOLVE

5)



Applying rule and putting these SMILES, we get :

